Object Detection in Large-Scale Remote-Sensing Images Based on Time-Frequency Analysis and Feature Optimization

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***Abstract*— Recently, optical remote-sensing images have been steadily growing in size, as they contain massive data and complex backgrounds. This trend presents several problems for object detection, for example, increased computation time and memory consumption and more false positives due to the complex backgrounds of large-scale images. Inspired by deep neural networks combined with time-frequency analysis, we propose a time-frequency analysis-based object detection method for large-scale remote-sensing images with complex backgrounds. We utilize wavelet decomposition to carry out a time-frequency transform and then integrate it with deep learning in feature optimization. To effectively capture the time-frequency features, we propose a feature optimization method based on deep reinforcement learning to select the dominant time-frequency channels. Furthermore, we design a discrete wavelet multiscale attention mechanism (DW-MAM), enabling the detector to concentrate on the object area rather than the background. Extensive experiments show that the proposed method of learning from time-frequency channels not only solves the challenges of large-scale and complex backgrounds, but also improves the performance compared to the original state-of-the-art object detection methods. In addition, the proposed method can be used with almost all object detection neural networks, regardless of whether they are anchor-based or anchor-free detectors, horizontal or rotation detectors.**

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I. Introduction

REMOTE-sensing image processing and analysis play an increasingly significant role in earth observation [1]–[3]. Object detection is a fundamental but challenging task in the remote-sensing community and plays an important role in wide-scope applications including target reconnaissance, land monitoring, and environmental conservation [4]–[7]. For instance, object detection in the remote-sensing system can monitor airports and harbors to obtain the spatial distribution and geographic coordinates of objects such as aircraft and ships [8]–[11]; it can also be used to observe the growth of crops [12]–[14].

Recently, the scale of remote-sensing images has been growing along with the development and maturity of key materials and devices [15], such as high-performance silicon carbide and high-stability carbon fiber. As shown in Fig. 1, a remote-sensing image with a size of 12 000 × 5000 pixels contains massive amounts of data, for example, many objects that are distributed in a 54.6 km^2 area. Large-scale images, such as Fig. 1, present several problems for object detection. Large-scale images not only have large image size, but also involve large number of images [16]. First, when the scale of the input image is excessively large, the computation time and memory consumption increase quadratically, making the object detection operate slowly on the current hardware. Second, massive and complex backgrounds may appear in a real-world scenario, which would introduce more false positives. For instance, in an ocean area with undulating waves, rising waves may be incorrectly detected as ships; in crowded parking lots, vehicles in the shade of trees may be neglected.

To resolve the problems in large-scale remote-sensing images with complex backgrounds, recent efforts [17]–[21] have detected objects using time-frequency analysis methods. In the process of time-frequency analysis, the input large-scale remote-sensing images are downsampled after being filtered, and hence, the sizes of these images can be reduced [22], [23]. In addition, this type of method offers effective multiscale and multidirection decomposition for image processing [24]–[27],

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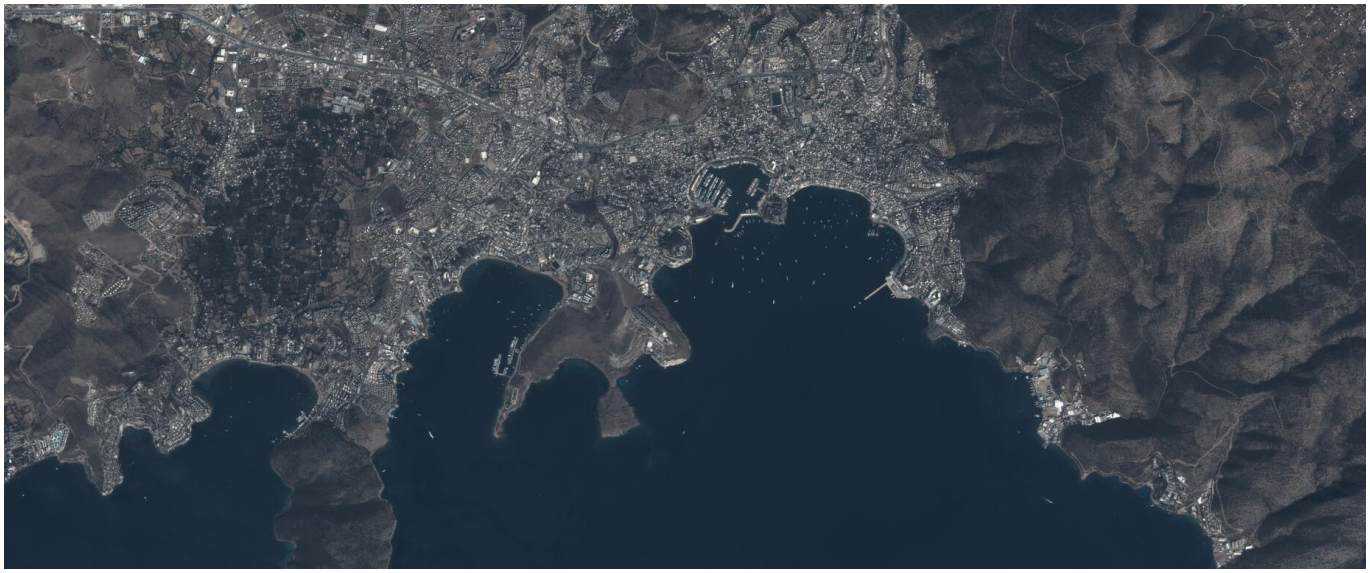


Fig. 1. This remote-sensing image was shot by the JiLin No. 1 business satellite. The objects in the image are located in Bodrum, Turkey. The size of this image is 12 000 × 5000 pixels, and the area covers 54.6 km^2. The spatial resolution of this image is 0.91 m.

and the direction and frequency characteristics of the backgrounds are quite different from the objects. These features motivate us to use time-frequency analysis to realize the deep representation and information mining of object features for large-scale images, thereby addressing the challenges stemming from massive data and complex backgrounds.

Despite the potential benefits of time-frequency transforms, it is not straightforward to apply a time-frequency analysis-based method to an object detection problem since two major technical challenges remain.

1) We must consider how to select the dominant time-frequency channels from the all time-frequency channels. After the time-frequency transform, too many time-frequency channels are generated. For instance, in the wavelet transform, there are 30 output channels after the transform with the input of an image consisting of R, G, and B color channels using a three-level wavelet decomposition. Among these time-frequency channels, few of them can be used by deep neural networks to generate accurate predictions, and these channels are referred to as the dominant time-frequency channels [28]. Since all of the time-frequency channels share similar distributions [29], it is challenging to identify the dominant time-frequency channels.
2) We must weaken the interference from complex backgrounds. In large-scale remote-sensing images, the objects and complex backgrounds usually overlap; hence, they are not easy to separate from each other. This problem inevitably causes interference in object detection for large-scale images.

For example, when an airplane is parked next to an airport bridge, ground service vehicles randomly park next to the airplane, which interferes with the detection of such airplanes. As a result, it is a challenging task to eliminate or even mitigate the interference from complex backgrounds.

In this article, we propose an object detection method for large-scale optical remote-sensing images with complex backgrounds. Our method learns from time-frequency analysis-based methods. We utilize wavelet decomposition to carry out a time-frequency transform and integrate it with deep learning in the feature optimization. Note that many channels are produced after applying a time-frequency transform to the image. With the aim of effectively capturing the time-frequency features, we propose a feature optimization method based on deep reinforcement learning to select the dominant time-frequency channels from the entire time-frequency channels. This procedure reduces computation complexity and ensures strong detection performance. In addition, we propose a discrete wavelet multiscale attention mechanism (DW-MAM) that enables the detector to pay more attention to the object area than the background. The proposed DW-MAM is derived from the wavelet function and can inherently extract the multiscale and multidirection features efficiently, which helps improve the mean average precision (mAP). The contributions of this article are summarized as follows.

1) We realize the feature optimization by leveraging deep reinforcement learning. In the game of feature optimization, the agent makes a choice with time-frequency channels after receiving the new state, and then, the environment evaluates the current mAP and feeds back the reward to the agent. Through the interaction between the agent and the environment, the agent manages to select the dominant time-frequency channels from all of the time-frequency channels.
2) We propose a novel attention mechanism DW-MAM and mathematically prove that DW-MAM is a special case of the attention mechanism. In addition, we integrate the proposed attention mechanism into the object detection network and achieve a better mAP.

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3) We conduct many experiments based on two widely used datasets, namely DOTA [30] (including DOTA-v1.0, DOTA-v1.5, and DOTA-v2.0) and DIOR [31]. The experimental results demonstrate that the state-of-the-art (SOTA) methods modified by our proposed time-frequency analysis is superior to the original SOTA methods, and the average performance is improved by 2%.

The remainder of this article is organized as follows. Section II describes the related works, and Section III presents our methodology. Then, Section IV presents the experimental results on the DOTA dataset (including DOTA-v1.0, DOTA-v1.5, and DOTA-v2.0) and DIOR data set. Finally, Section V concludes the article.

II. Related Works

At present, object detection is an active research field in computer vision and has also been widely explored in the field of remote sensing. In this section, we provide a survey of object detection related to remote sensing. Furthermore, we analyze some works on time-frequency analysis-based methods.

A. Object Detection

In the past years, inspired by the great progress of deep learning-based object detection methods in natural scene images, target detection in object remote-sensing images has been extensively studied to detect various geospatial objects. Similar to the area of natural scene images, the existing deep-learning-based remote-sensing image object detection methods can be generally categorized into three approaches: two-stage, single-stage, and anchor-free detectors.

1) Two-Stage Detectors: The region-based CNN (R-CNN) proposed by Girshick *et al.* [32] is one of the best known of the various region-based proposal approaches, which transform the detection problem into a classification and regression problem by extracting a set of object proposals (object candidate boxes).To obtain the precise localization of geospatial objects in high-resolution earth observation images, several research efforts to extend the framework of R-CNN to the field of earth observation. Long *et al.* [33] presented an unsupervised score-based bounding box regression (USB-BBR) method using the R-CNN framework.

Objects in remote-sensing images are often arbitrarily oriented due to the bird's-eye view. In order to solve the problems caused by the orientation variations of geospatial objects, Cheng *et al.* [34] proposed to learn a rotation-invariant CNN (RICNN) model in R-CNN framework used for multiclass geospatial object detection. Liu *et al.* [35] proposed a rotation R-CNN (RRCNN), which uses rotation region-of-interest (RRoI) pooling to extract rotation-invariant region features.

To overcome the problem of redundant feature computation on a large number of overlapping proposals in R-CNN, Ren *et al.* [36] proposed a two-stage object detection neural network called Faster R-CNN. This network is a very classic object detection network that consists of a backbone, a region proposal network (RPN), and a detection head. Although there are many novel two-stage and anchor-based detectors, the studies into and improvements of Faster R-CNN have never ceased. In this article, we also undertake extensive experiments on it. In the field of earth observation, Li *et al.* [5] presented a rotation-insensitive RPN by adding multiangle anchors into the existing RPN based on Faster R-CNN pipeline. Furthermore, in order to tackle the appearance ambiguity problem, they proposed a double-channel feature fusion network that can learn local and contextual properties along two independent pathways.

Compared with natural images, scale variations are larger in remote-sensing images. To circumvent issues caused by the size variations of geospatial objects, feature pyramids and image pyramids are widely used. Tayara and Chong [37] proposed a tightly connected feature pyramid network (FPN) through which high-quality multiscale semantic features can be generated, which can extract scale-invariant features. In order to further enhance the detection accuracy and speed, Wang *et al.* [6] proposed an end-to-end multiscale visual attention network (MS-VANS) that uses a jump-connected encoder–decoder to extract multiscale features from an entire image to detect objects with different scales. For each feature map at each scale, the detector learns a visual attention representation and then highlights the features of the regions by classifying and regressing branches, which greatly suppresses the clutter of the background and the occlusions of geospatial objects.

2) One-Stage Detectors: The two-stage detection method can achieve high accuracy because of the existence of a secondary correction mechanism. To further dramatically improve the efficiency of the algorithm and even achieve real-time object detection, various one-stage methods have been proposed, which detect the object through the regular and intensive sampling of its position, scale, and aspect ratio. YOLO proposed by Redmon *et al.* [38] was the first one-stage detector, which accomplishes the object detection through a regression strategy. Another classical one-stage method called single shot multibox detector (SSD) [39] generates a series of prior boxes with different scales and aspect ratios across multiple feature maps and predicts the category of each box.

Although this process greatly improves computational efficiency, the detection accuracy is slightly lower. Actually, the balance between precision and speed is a challenging topic for object detection methods in large-scale remote-sensing images. Many scholars have undertaken deep investigations into the one-stage remote-sensing image object detection algorithm and gradually approached the performance of the two-stage detection model. Driven by the excellent performance of YOLO and SSD in natural scene image object detection, Liu *et al.* [9] designed a framework for detecting arbitrary-oriented ships. This model can directly predict rotated/oriented bounding boxes (OBBs) by using YOLO architecture as the fundamental network. Besides, Wang *et al.* [16] proposed the feature-merged single-shot detection (FMSSD) network, which uses an atrous spatial feature pyramid (ASFP) to integrate the context information of multiscale features, and they proposed an area-weighted

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loss function to make the network focus more attention on small targets. Du *et al.* [40] proposed the saliency-guided loss function on the basis of the single-shot detection network RetinaNet [41], which treats the training samples differently based on input complexity to avoid the excessive contribution of simple samples in the training process.

Wu *et al.* [20] proposed the ORSIm detector, which is integrated with multichannel feature extraction, feature learning, rapid image pyramid matching, and a strengthening strategy. This detector adopted a new spatial frequency channel characteristic (SFCF) by considering the rotation-invariant channel in the frequency domain structure characteristic and the space of the original channel characteristics. The goal was to obtain better high-level semantic features for rotating objects with a large aspect ratio, dense distribution, and extremely unbalanced categories. Lei *et al.* [42] introduced a significance constraint into the convolutional neural network (CNN) model to guide the model training and proposed the region-enhanced convolutional neural network (RECNN). To further enhance the feature mapping of object regions, these authors introduced a multilevel fusion strategy to fuse convolutional layers with different resolutions.

3) Anchor-Free Detectors: An anchor-based method that is based on a predefined bounding box requires a manual design of the anchor parameters according to the size of the object, which requires tedious and heuristic configurations. Indeed, a large number of anchors causes redundancy in calculation. Because the anchor-free method represented by FCOS [43] eliminates the redundant computations caused by anchors, it enables object detection to be more efficient in real time. There is also some exploratory work that uses anchor-free detectors in remote sensing.

To alleviate the channel imbalance between features at different levels in the process of feature fusion, Zhang *et al.* [44] proposed an anchor-free rotation ship detector GRS-Det that extracts multiscale features through a feature extraction network based on a selective cascade module. In addition, these authors designed a rotating Gaussian mask to solve the mislabeling problem of the rotating bounding boxes. Zhou *et al.* [45] introduced the polar coordinate system into object detectors and proposed the polar anchor-free object detector P-RSDet for remote-sensing images. This detector takes the center point of each object as the pole and the positive horizontal direction as the pole axis to establish the polar coordinate system. In this coordinate system, by predicting the center point, P-RSDet regresses a polar radius and two polar angles to realize the detection of objects with arbitrary direction. Yu *et al.* [46] proposed a novel one-stage anchor-free capsule network (OA-CapsNet) for detecting geospatial objects from remote-sensing images. By employing a capsule FPN architecture as the backbone, and designing a centerness-assisted one-stage anchor-free object detection strategy, the proposed OA-CapsNet performs effectively in recognizing arbitrarily orientated and diverse-scale geospatial objects.

B. Time-Frequency Analysis-Based Methods

Time-frequency analysis-based methods have achieved promising performance in digital signal processing and image processing. In recent years, there have been some articles combining frequency domain analysis with deep neural networks. Torfason *et al.* [47] performed image-understanding tasks such as classification and segmentation directly on the compressed representation generated by the compression method. This work shows that the proposed method greatly reduces the computational cost and ensures the performance at the same time. Wu *et al.* [48] trained a deep network for video action recognition directly on compressed video formats such as H.264 and MPEG-4. Their proposed method is not only superior to other methods, but also halves the training time. Gueguen *et al.* [23] trained a CNN directly on images after they were transformed by a discrete cosine transform (DCT). These authors showed that their proposed method obtained similar performance with half of the time cost.

However, frequency transform methods have a fatal flaw when facing a nonstationary signal, which cannot distinguish nonstationary signals whose frequency varies with time. To address this issue, time-frequency analysis [49] was proposed. In this article, we utilize the time-frequency analysis-based method, which concentrates on multiscale and multidirection features. By exploiting the expressive power of time-frequency analysis, the proposed method not only solves the challenges of large-scale and complex backgrounds, but also improves performance.

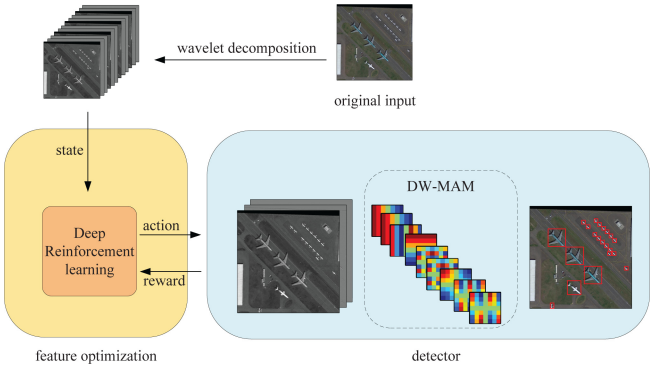


Fig. 2. Framework of the proposed method. The deep reinforcement learning network in the figure selects a limited number of time-frequency channels with the input of the original RGB images. The detector utilizes the selected time-frequency channels to obtain the detection result and outputs the mAP as the reward of the deep reinforcement learning network.

III. Methodology

In this section, we describe in detail the proposed method of learning from time-frequency features for optical remote-sensing images. The framework of the proposed method is shown in Fig. 2. We describe the proposed method according to the following four parts: 1) wavelet decomposition; 2) feature optimization; 3) DW-MAM; and 4) modifications to detectors.

A. Wavelet Decomposition

To obtain the time-frequency channels using wavelet decomposition, we first split the original RGB images into R, G, and B color channels. Then, these channels are decomposed, which is shown in Fig. 3. We take the R color channel as an example to show the process of wavelet decomposition. The G and B

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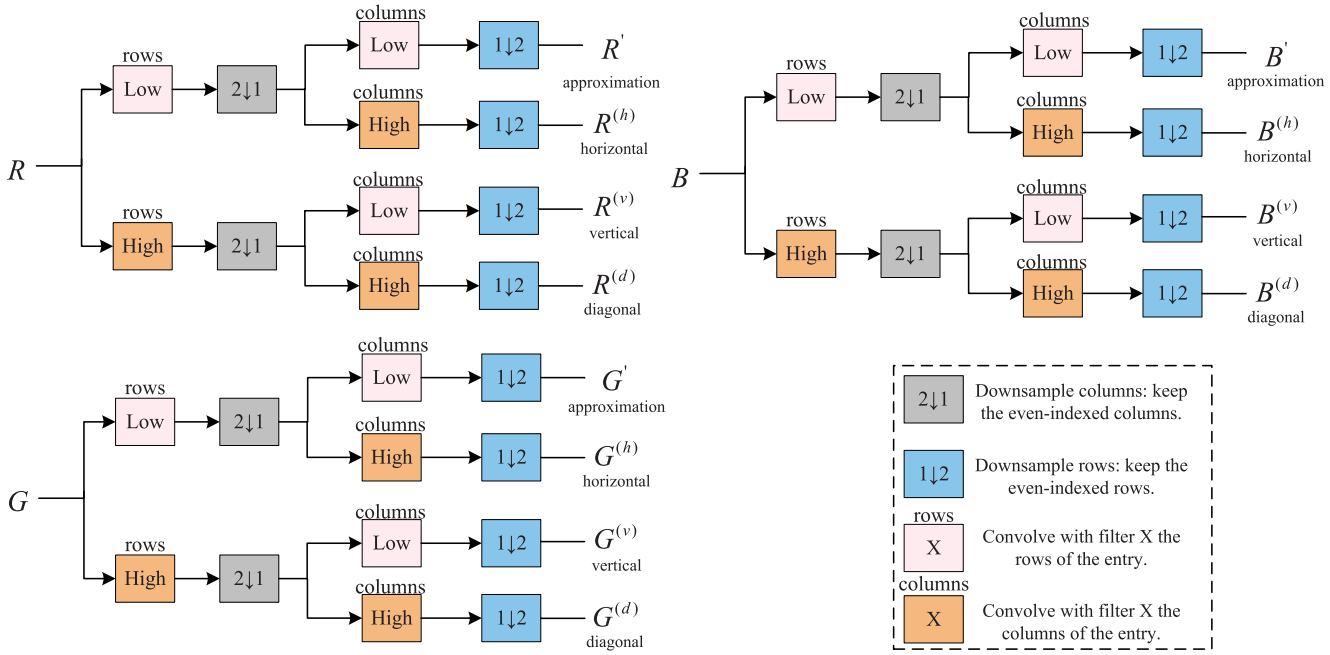


Fig. 3. Decomposition process of the R, G, and B color channels. Each color channel is decomposed into four components: the approximation time-frequency channel, the horizontal orientation's time-frequency channel, the vertical orientation's time-frequency channel, and the diagonal orientation's time-frequency channel.

color channels are processed in the same way as the R color channel.

To derive the approximation subband, the R color channel is convolved with a low-pass filter in rows, and we retain only the even-indexed columns. The result is convolved with a low-pass filter in the columns, and we retain only the even-indexed rows. At this point, the approximation subband R' is obtained.

To derive the horizontal subband, the R color channel is convolved with a low-pass filter in rows. Here, we only retain the even-indexed columns. The result is convolved with a high-pass filter in columns, and we only save the even-indexed rows. Thus, the horizontal subband $R^{(h)}$ is obtained.

For the vertical subband, the R color channel is convolved with a high-pass filter in rows. Then, we only retain the even-indexed columns. The result of this process is convolved with a low-pass filter in columns, and we only save the even-indexed rows. Then, the vertical subband $R^{(v)}$ is obtained.

Last, to obtain the diagonal subband, the R color channel is convolved with a high-pass filter in rows, and we only keep the even-indexed columns. The result is convolved with a high-pass filter in columns, and we only retain the even-indexed rows. Hence, the diagonal subband $R^{(d)}$ is obtained.

As shown in Fig. 4, we split the original RGB images into R, G, and B color channels. Then, we carry out 2-D wavelet decomposition for the R, G, and B color channels. The level J of the 2-D wavelet decomposition in this article is set to 1. One color channel can generate an approximation and three orientations. In this article, the approximation and the details of the three orientations are all considered time-frequency channels. In other words, we obtain 12 time-frequency channels in total, which are concatenated together to form a tensor cube. Finally, every time-frequency channel is normalized by the mean and variance calculated from the data set. According to the order of the time-frequency channels' spatial position in Fig. 4, we number these channels from 0 to 11 from left to right and from top to bottom.

The shape of the original RGB image is $H \times W \times 3$. After the wavelet decomposition, the shape of the tensor cube is $(H/2) \times (W/2) \times 12$. If we upsample the original RGB image to the shape of $2H \times 2W \times 3$ in advance, then the shape of the tensor cube will be $H \times W \times 12$.

B. Feature Optimization

Among the 12 time-frequency channels that have been generated, we find through experiments that only dominant time-frequency channels can be used by deep neural networks to generate accurate predictions. Consequently, the strategy is to select the dominant time-frequency channels from all of the time-frequency channels. This process is called feature optimization.

Reinforcement learning [50] is good at learning ideal policies for decision issues. Q-learning [51] is one of the most popular reinforcement learning algorithms. In recent years, deep Q-networks (DQNs) have implemented Q-learning via a flexible deep neural network. This network has been evaluated on 2600 Atari games, and it reached human-level performance on many games. The double DQN (DDQN) is a novel algorithm with far greater capabilities than DQN that not only yields more accurate predictions but also leads to higher scores on Atari games.

The purpose of DDQN is to allow an agent to learn a policy: $\pi(s, a)$. The policy can decide to operate the action

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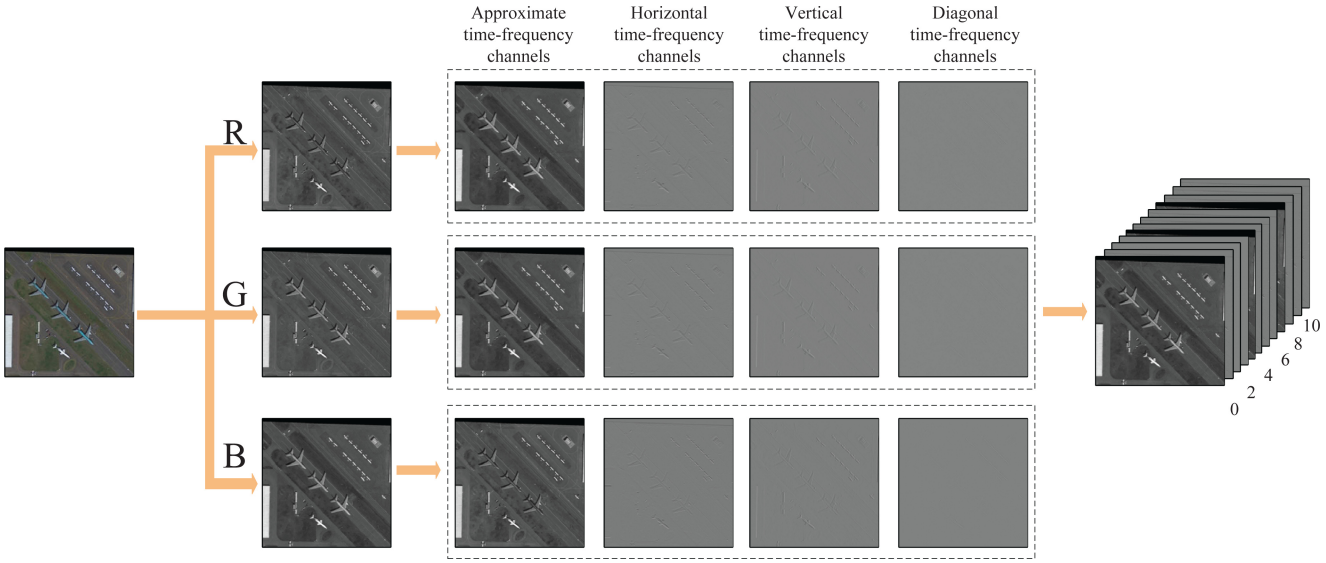


Fig. 4. Illustration of 12 time-frequency channels after a wavelet decomposition.

a in the given state s. The best policy is able to utilize the rewards r provided by the environment to maximize the future accumulated rewards for actions in a Markov decision process (MDP). This policy relies on the initial and final states as well as the reward. To convey the future accumulated reward, the state-action value function, which is called Q-function, is given by the following equation:

$$Q_\pi(s_t, a_t) = \mathbf{E}_\pi\left[r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \cdots\right] \tag{1}$$

where a_t is the action at time t, s_t is the state at time t, γ is the discount factor, and $\gamma \in (0, 1)$.

The state-action value function Q satisfies the Bellman equation given by the following equation:

$$Q(s_t, a_t) = r_t + \gamma\, Q'\left(s_{t+1}, \arg\max_a Q(s_{t+1}, a)\right) \tag{2}$$

where s_{t+1} is the state at time $t+1$, Q represents the Q-function of the evaluation network in DDQN, and Q' represents the Q-function of the target network in DDQN.

We utilize DDQN to select the dominant time-frequency channels from all of the time-frequency channels. We consider object detection as the environment, the generated time-frequency channels as the state, the time-frequency channel selection as the action, and the mAP of our detector as the reward. As such, in the game of feature optimization, the agent makes a choice in terms of the time-frequency channels after receiving the new state, and then, the environment evaluates the current mAP and feeds back the reward to the agent. Through the interaction between the agent and the environment, the maximum cumulative reward value can be obtained so that the agent manages to select the dominant time-frequency channels from all of the time-frequency channels.

However, it is worth noting that DDQN is better at handling a limited action space rather than an unlimited or large action space. In the original DDQN, the output of the network is determined by the action space, such as "left," "right," or "fire" in Atari games. In this case, the output layer consists of three neurons. Selecting the dominant time-frequency channels from all of the time-frequency channels is a combinatorial problem. There are $C_n^k = (n!/(n-k)!*k!)$ different selections if we select k dominant time-frequency channels from n total time-frequency channels. For example, if we select three dominant time-frequency channels from 12 time-frequency channels generated by wavelet decomposition, then there are 220 different selections. Thus, the dimension of this action space is 220, which is too large to be processed by DDQN. It is difficult to imagine that DDQN would output 220 independent state-action value functions $Q(s, a)$.

To tackle this intricate problem, we put forward a novel solution that only involves minor changes to the original DDQN: we replace the neurons with a matrix. This matrix is defined as matrix A with the shape of $C \times c$, where C represents the number of total subbands and c represents the number of the selected time-frequency channels ($c \in (0, C)$). We find the maximum value in the first axis of A. Then, we obtain a vector y containing c elements that represents the selected time-frequency channels.

The state in the proposed method is defined as matrix S with the shape of $H \times W \times 12$. The reward function in this article is defined as follows:

$$r_t = F - f \tag{3}$$

where F represents the mAP of the object detection result at time t, and f is a hyperparameter [$f \in (0, 1)$].

Due to the above changes, the state-action value function has to be changed from (2) to the following:

$$Q(s_t, \upsilon, c) = r_t + \gamma\, Q'\left(s_{t+1}, \arg\max_\upsilon Q(s_{t+1}, \upsilon, c), c\right) \tag{4}$$

where υ is the maximum value in $A[:, n]$.

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The training process is presented in Algorithm 1. DDQN is updated by the following equation:

$$\theta \leftarrow \theta - \alpha_t \cdot \frac{1}{n}\sum_{i=1}^{n}[Y_i - Q_\theta(s_i, a_i)] \cdot \nabla_\theta Q_\theta(s_i, a_i). \quad (5)$$

Algorithm 1 Training Process of DDQN

Input: Replay memory $\mathcal{M}$; number of iterations T; batch size n; exploration probability $\epsilon \in (0, 1)$; Q-function Q of deep Q-networks for evaluation; Q-function Q' of deep Q-networks for target; integer T_{target} to update the target network Q'.

Output: Q-function Q_θ of deep Q-networks for evaluation; Q-function Q'_θ of deep Q-networks for target.

Initialize replay memory $\mathcal{M}$ so that it is empty;
Initialize Q_θ and Q'_θ with random weights θ;
Initialize initial state s_0 of DDQN.
for $t = 0, 1, \ldots, T$ **do**
  Given state s_t, generate a random number ξ, $\xi \in (0, 1)$;
  if $\epsilon \leq \xi$ **then**
    Take action a_t according to Q;
  else
    Take action a_t in random;
  end if
  Obtain reward r_t, and reach new state s_{t+1};
  Store transition (s_t, a_t, r_t, s_{t+1}) in $\mathcal{M}$;
  if $n \leq t$ **then**
    Sample a batch of (s_i, a_i, r_i, s_{i+1}) from $\mathcal{M}$, $i \in [0, n-1]$;
    Target $y_i = r_i + \max_a Q'(s_{i+1}, a)$, $i \in [0, n-1]$;
    Update the parameters of Q_θ to let $Q(s_i, a_i)$ converge to y_i, $i \in [0, n-1]$;
  end if
  Set every T_{target} to make $Q'_\theta = Q_\theta$.
end for

In our proposed method, DDQN has been modified according to our unique design. The input is the original RGB image and the output is the selection result of the time-frequency channels. Specifically, the network structure consists of ResNet-50 [52]. After ResNet-50, there is a maximum pooling layer and a fully connected layer, which outputs a tensor with the shape of $C \times c$.

As shown in Fig. 5, if we select three time-frequency channels from 12 time-frequency channels, then the fully connected layer outputs the tensor with the shape of 12×3.

C. DW-MAM

Interference cannot be eliminated in large-scale remote-sensing images with complex backgrounds, and it causes many false detection results. As a result, it is quite a challenge to weaken the interference from complex backgrounds. We utilize the wavelet function to derive a novel attention mechanism that concentrates on multiscale and multidirection features and eliminates the interference from complex backgrounds. In addition, we integrate this novel attention mechanism into the existing object detection network and achieve better mAP.

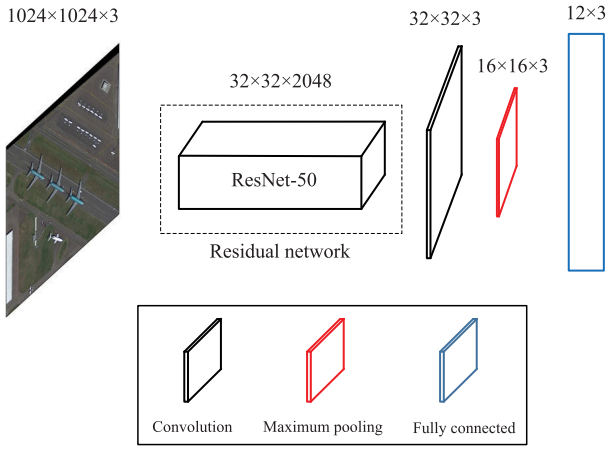


Fig. 5. Network structure of the modified DDQN. It consists of ResNet-50 [52], a maximum pooling layer, and a fully connected layer.

1) Attention Mechanism: According to [53]–[55], the attention mechanism can be written as follows:

$$\text{attention} = \text{sigmoid}(\text{map}(\text{gap}(X))) \quad (6)$$

where $X \in \mathbb{R}^{C' \times H' \times W'}$ is the tensor in the feature extraction network, C' is the number of channels, H' is the height of the tensor, W' is the width of the tensor, attention $\in \mathbb{R}^{C'}$ is the weighted tensor, sigmoid is the sigmoid function, map is the mapping operation (such as the fully connected layer and 1×1 convolution), and gap is the global average pooling.

If we decompose the attention mechanism channel by channel, then we can obtain the following expression:

$$\tilde{X}_{:,i,:,:} = \text{attention}_i X_{:,i,:,:} \quad (7)$$

where $i \in \{0, 1, \ldots, C' - 1\}$, $\tilde{X}$ is the result of the attention mechanism, attention_i is the ith element in the attention vector, and $X_{:,i,:,:}$ is the ith channel of the input tensor.

2) Theoretical Analysis of DW-MAM: The 2-D discrete wavelet transform is composed of a 2-D scaling function and three 2-D wavelet functions. The 2-D scale function corresponds to the approximate time-frequency channel, the first 2-D wavelet function corresponds to the horizontal time-frequency channel, the second 2-D wavelet function corresponds to the vertical time-frequency channel, and the third 2-D wavelet function corresponds to the diagonal time-frequency channel. We employ the 2-D scaling function to construct DW-MAM, which is defined by the following formulas:

$$W_\varphi(p, m, n) = \frac{1}{\sqrt{MN}}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x, y)\varphi_{p,m,n}(x, y) \quad (8)$$

$$\varphi_{p,m,n}(x, y) = 2^{\frac{p}{2}}\varphi\left(2^p x - m, 2^p \text{y-n}\right) \quad (9)$$

$$\varphi(x, y) = \varphi(x)\varphi(y) \quad (10)$$

where $p \in \{0, 1, \ldots, P - 1\}$, P is the number of scales in the 2-D wavelet, and $P = \log_2 M$. In addition, M is the height of the input image, N is the width of the input image, $m \in [0, 2^p]$, $n \in [0, 2^p]$, and $\varphi(x)$ is the scaling function according to the type of wavelet.

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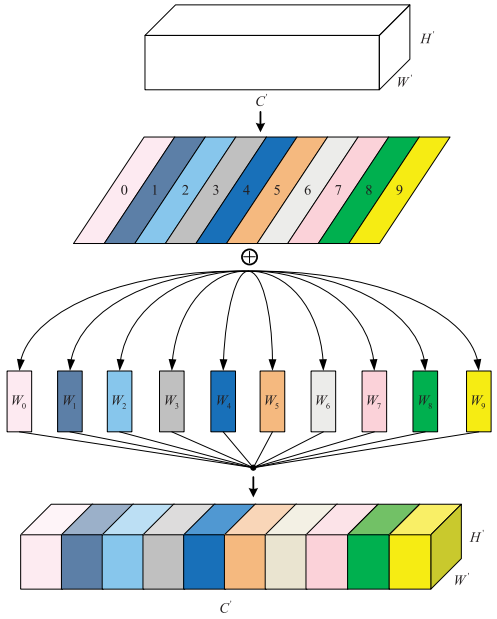


Fig. 6. Illustration of DW-MAM. The tensor is cut into ten blocks. There are ten coefficient matrices, which are represented by W_i in the figure. After elementwise multiplication and summation, we obtain the final result of DW-MAM.

By substituting (8)–(10) into the calculation and simplifying, we can derive the following expression:

$$T = WF \tag{11}$$

where W represents the wavelet transforms, F represents the input tensor, and T is the result of the calculation. Obviously, this equation conforms to the expression of (7), which proves that the proposed method is a qualified attention mechanism.

3) Constructing DW-MAM: In the construction process of DW-MAM, the $H' \times W' \times C'$ tensor in the feature extraction network or in the backbones of the detectors is cut into P blocks. The shape of each block is $H' \times W' \times (C'/P)$. We use (8) to calculate P coefficient matrixes, which are essentially the values of discrete wavelet scaling functions; they are also numbered from $0, 1, \ldots, P-1$ in order. Because of $m \in [0, 2^p]$ and $n \in [0, 2^p]$, the size of each coefficient matrix is different. We interpolate all of the coefficient matrices to the same size of $H' \times W'$. Then, each block and coefficient matrix is multiplied according to the number. By elementwise multiplication and summation, the feature extraction network with DW-MAM defined by (8)–(10) is implemented.

As shown in Fig. 6, $P = 10$ since $H' = W' = 1024$. The tensor is cut into ten blocks. There are ten coefficient matrices, which are represented by W_i in the figure. After elementwise multiplication and summation, we obtain the final result of DW-MAM.

D. Modifications to Detectors

After the wavelet decomposition and feature optimization, the shape of the obtained tensor is not suitable for the original object detection detector. The tensor cube of $(H/2) \times (W/2) \times c$ cannot be fed into the detector due to the mismatch of the shape. We need to remove the stem convolution layer in ResNet [52] and reserve all of the remaining layers. In the following layer, the number of channels is modified to fit the channel of the selected time-frequency channels c. The modifications to detectors are shown in Fig. 7. In this way, the wavelet decomposition is considered as a basic feature extraction convolution layer.

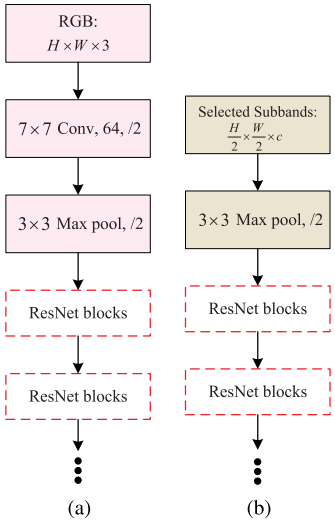


Fig. 7. Comparison of the original and modified backbones. (a) Original backbone of Faster R-CNN. (b) Modified backbone of Faster R-CNN.

We demonstrate that the proposed method based on wavelet transforms can be applied to all of the existing object detection networks that were designed for RGB images with minor changes. Moreover, the modified detectors have less parameter and computational costs than the original ones.

In the proposed method, we supply static feature optimization and dynamic feature optimization. A static feature optimization method has the advantage of less computational cost, while a dynamic feature optimization method can achieve better performance.

1) Static Feature Optimization: First, we utilize the whole training data set to train the DDQN and the detector. The DDQN and the detector share the same RGB images batch. The inference results of DDQN (the selected wavelet time-frequency channels) are sent to the detectors, which are also the inputs of the detectors. Second, the detector carries out forward and backward propagation. The mAPs are sent to the DDQN as the reward. Finally, we find the time-frequency channels with the most occurrences, which are considered as the static time-frequency channels.

We utilize these static time-frequency channels to train the modified detector from scratch. After training, the trained detector without DDQN inferences on the testing dataset and obtains the final mAP.

2) Dynamic Feature Optimization: The dynamic feature optimization method also follows the above training process of DDQN and the detector. However, on the testing dataset, the DDQN works with the detector. That is to say, the DDQN makes the time-frequency channel selection according to the current input image batch in real-time. The selection result of the time-frequency channels are changed in every batch. In this way, we obtain the final mAP.

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IV. Experimental Results and Analysis

To evaluate the performance of the proposed method in this section, we conduct experiments on the DOTA dataset [30] (including DOTA-v1.0, DOTA-v1.5, and DOTA-v2.0) and DIOR dataset [31]. In addition, we compare the SOTA methods modified by our time-frequency analysis method with the original SOTA methods.

A. Dataset

The DOTA dataset [30] is a vital dataset for object detection in remote sensing, including 15 different categories, such as planes, ships, storage tanks, baseball diamonds, harbors, and bridges. There are 2806 large-scale optical remote-sensing images in DOTA-v1.0 that were collected from Google Earth, the JiLin No. 1 business satellite, the GaoFen No. 2 satellite, and the China Centre for Resources Satellite Data and Application. The resolutions of the images range from 800 × 800 to 4000 × 4000. In addition to their diversity, the images in DOTA have various scales, orientations, and shapes. To compare the proposed method with conventional detectors, we split all of the original images in DOTA to images of size 1024 × 1024. After the splitting process, there are 21 046 images in the training dataset and 10 833 images in the testing dataset of DOTA-v1.0. For the DOTA dataset, there are two object detection tasks: detection with horizontal bounding boxes (HBBs) and detection with OBBs.

The DIOR dataset [31] consists of 23 463 images covered by 20 object categories and each category contains about 1200 images. DIOR dataset holds large variations because the images are captured in different imaging conditions, weathers, and seasons. Besides, objects in DIOR possess high inter-class similarity and intraclass diversity. However, objects in DIOR are both annotated by HBBs, which are not suitable for detectors based on OBBs.

The proposed method is implemented by mmdetection [56], which is an open source object detection toolbox based on PyTorch.

All of the experiments are trained and tested on one NVIDIA GeForce GTX 1080 with 8 GB GPU memory. Due to the limited GPU memory, the batch size is set to 2 in our proposed method.

During training, we utilize the network weights pretrained on ImageNet to speed up the convergence. Specifically, we utilize ResNet-50 pretrained weights to initialize the backbone of the detector. In our proposed method, the stochastic gradient descent (SGD) is utilized to optimize our network. Initially, the learning rate is set to 0.01, the momentum is set to 0.9 and the weight decay is set to 10^{-4}. For every epoch, the learning rate is multiplied by 0.9, and every network is trained by 24 epochs in total.

In object detection, mAP is the most widely used evaluation metric and evaluation indicator. As a result, it is regarded as a reward for reinforcement learning in our proposed method. In addition, we utilize mAP to compare the proposed method to other advanced methods.

To compare the proposed method based on wavelet transforms to the original detector, we utilize Faster R-CNN as a baseline. In [30], it is reported that Faster R-CNN obtains 60.46 mAP when trained on 1024 × 1024 patches of DOTA.

In our article, we utilize the official implementation of Faster R-CNN, which is based on mmdetection. The input of it is the original RGB image with the shape of 1024 × 1024. We ultimately obtain 63.1 mAP. When we utilize a conventional downsampling operation, the original RGB 1024 × 1024 images are converted to 512 × 512. When we utilize these downsampled images to train Faster R-CNN, we obtain 57.5 mAP, which presents a noticeable decline. As a result, we regard 63.1 and 57.5 as our baselines for 1024 × 1024 and 512 × 512 images, respectively.

B. Ablation Study

To further illustrate the improvements of the key component in the proposed method, we carry out a series of experiments. The key components in the proposed method are time-frequency analysis, DW-MAM, and static/dynamic feature optimization. In view of these components, we conduct the experiments separately.

It is known that wavelet decomposition has a downsampling effect. Therefore, we upsample the original RGB images before transforming. As a result, even after the wavelet decomposition, the inputs of our proposed method are images with the width and height of 1024. Note that all of the experiments in the ablation study are conducted under the condition that the upsampling operation is enabled before the wavelet decomposition.

The point needs to be emphasized here, with dynamic feature optimization and DW-MAM, the proposed method exhibits promising performance, which is also the highest mAP in our article. When DW-MAM and dynamic DDQN feature optimization are enabled, we call it optimal configuration (abbreviated as optimal). As for why DW-MAM and Dynamic DDQN work, we will discuss it later.

1) Ablation Study for Feature Optimization: To demonstrate the superiority of the feature optimization based on DDQN, we compare three different combinations of time-frequency channels, which correspond to three rows in Table I. The first row shows the performance difference using manually selected 12, 9, and 6 time-frequency channels as inputs.

From the first row of Table I, we can see that when we utilize part of the time-frequency channels, the obtained result is inconsistent with our expectations. The algorithm's performance is not as good as we expected and, in fact, rather poor. In a recent article on the interpretability of deep neural networks, Haohan Wang pointed out that deep neural networks tend to pay more attention to high-frequency information, and hence, the high-frequency components in the image determine the prediction result, while the low-frequency components greatly degrade performance [28]. The results of this experiment are consistent with the above findings.

In the second row of Table I, we compare the difference between using DDQN and manually selecting three time-frequency channels. The "−" means that we cannot get mAP after uploading the testing results to the official website. In other words, it cannot detect any objects. Comparing the

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TABLE I
COMPARISON WITH FEATURE OPTIMIZATION AND WITHOUT FEATURE OPTIMIZATION

input	time-frequency channels	time-frequency channels	time-frequency channels
shape	$1024 \times 1024 \times 12$	$1024 \times 1024 \times 9$	$1024 \times 1024 \times 6$
DW-MAM	$\surd$	$\surd$	$\surd$
feature optimization	0,1,2,3,4,5,6,7,8,9,10,11	1,2,3,5,6,7,9,10,11	0,2,4,6,8,10
static/dynamic	static	static	static
DOTA-v1.0 HBB mAP	10.8	55.5	24.1
DOTA-v1.5 HBB mAP	9.7	49.7	21.5
DOTA-v2.0 HBB mAP	7.1	36.6	15.8
DIOR	11.1	56.9	24.7
speed (fps)	12.6	13.2	13.7
input	time-frequency channels	time-frequency channels	time-frequency channels
shape	$1024 \times 1024 \times 3$	$1024 \times 1024 \times 3$	$1024 \times 1024 \times 3$
DW-MAM	$\surd$	$\surd$	$\surd$
feature optimization	DDQN	0,4,8	1,2,3
static/dynamic	static	static	static
DOTA-v1.0 HBB mAP	63.6	-	61.4
DOTA-v1.5 HBB mAP	56.7	-	54.6
DOTA-v2.0 HBB mAP	41.5	-	40.7
DIOR	65.0	-	63.1
speed (fps)	14.3	14.3	14.3
input	time-frequency channels	time-frequency channels	time-frequency channels
shape	$1024 \times 1024 \times 1$	$1024 \times 1024 \times 4$	$1024 \times 1024 \times 6$
DW-MAM	$\surd$	$\surd$	$\surd$
feature optimization	DDQN	DDQN	DDQN
static/dynamic	static	static	static
DOTA-v1.0 HBB mAP	59.5	63.2	58.6
DOTA-v1.5 HBB mAP	53.4	57.1	52.2
DOTA-v2.0 HBB mAP	39.1	41.6	38.0
DIOR	60.5	65.5	59.8
speed (fps)	16.3	14.1	13.7

first column with the third column, we can see that by selecting three time-frequency channels as input, the performance improves significantly compared to the previous selection of other number of channels, and the result using DDQN is better than that using manual selection.

In the last row of Table I, we compare the differences between the various numbers of time-frequency channels selected as inputs using DDQN, and we can notice that the best results are obtained using three channels.

This finding proves that simply using the time-frequency analysis method does not have any effect. The time-frequency analysis method works only if it is combined with the DDQN feature optimization proposed in this article.

2) Ablation Study for DW-MAM: To show the benefit of DW-MAM, we compare two different situations that have the same settings with the exception of whether or not DW-MAM is used. In Table II, we can observe that mAP has about 3% improvement on all four datasets after using DW-MAM, while other methods are kept consistent. We also note that the recognition speed of the algorithm decreases by only 0.1 fps after using DW-MAM, which means that the cost of computation increases slightly. The above results indicate that the performance when using DW-MAM is manifestly better than the case without DW-MAM.

3) Ablation Study for Static and Dynamic Feature Optimization: To show the benefit of dynamic feature optimization, we compare two different situations. They have the same settings, except for whether we use dynamic feature optimization or static feature optimization. In Table III, compared to static feature optimization, mAP improved from 65 to 66.1 on the DIOR dataset after using dynamic feature optimization, and even improved by about 2% on the three DOTA datasets. Therefore, the performance when using dynamic feature optimization is clearly better than that with static feature optimization. However, by using dynamic DDQN, the efficiency of the algorithm decreases, which can be seen from the reduction in fps from 14.3 to 7.6. The above experimental results show that dynamic DDQN feature optimization can achieve better mAP results and obtain better performance.

C. Comparison With SOTA Methods

To evaluate the proposed method, we compare it with other advanced methods such as DCTNet [17], densely coded labels (DCLs) [57], Gaussian Wasserstein distance (GWD) [58],

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TABLE II
COMPARISON OF THE METHOD BASED ON WHETHER DW-MAM IS USED

input	time-frequency channels	time-frequency channels
shape	$1024 \times 1024 \times 3$	$1024 \times 1024 \times 3$
DW-MAM	$\times$	$\surd$
feature optimization	DDQN	DDQN
static/dynamic	dynamic	dynamic
DOTA-v1.0 HBB mAP	61.9	65.1
DOTA-v1.5 HBB mAP	56.4	59.2
DOTA-v2.0 HBB mAP	42.1	44.3
DIOR	63.2	66.1
speed (fps)	7.7	7.6

TABLE III
COMPARISON OF STATIC FEATURE OPTIMIZATION AND DYNAMIC FEATURE OPTIMIZATION (MODIFIED ON FASTER R-CNN)

input	time-frequency channels	time-frequency channels
shape	$1024 \times 1024 \times 3$	$1024 \times 1024 \times 3$
DW-MAM	$\surd$	$\surd$
feature optimization	DDQN	DDQN
static/dynamic	dynamic	static
DOTA-v1.0 HBB mAP	65.1	63.6
DOTA-v1.5 HBB mAP	59.2	56.7
DOTA-v2.0 HBB mAP	44.3	41.5
DIOR	66.1	65.0
speed (fps)	7.6	14.3

and SCRDet [59]. What is more, we carry out extensive experiments on different datasets such as DOTA-v1.0, DOTA-v1.5, DOTA-v2.0, and DIOR.

1) Experiments With Our Proposed Method: As shown in Table IV, we utilize DCTNet-192 and DCTNet-24 on the DOTA dataset. DCTNet is a method based on frequency domain transformation. DCTNet-192 [17] represents that after the DCT transform, 192 channels of tensors with the shape of 56×56 are sent to Faster R-CNN. DCTNet-24 [17] represents that after the DCT transform, 24 channels of tensors with the shape of 56×56 are sent to Faster R-CNN. Take the DOTA-v1.0 dataset as an example, we obtain the results of 43.1 and 44.3 mAP, which are far less than the baseline in our article.

Faster R-CNN fed by the original RGB images with the shape of $1024 \times 1024 \times 3$ can obtain 63.1 mAP, which is the baseline in our article (as mentioned above). Because the width and height are 1024, this method is called Faster R-CNN-1024 in Table IV. However, when we utilize conventional downsampling operations, the input of Faster R-CNN is converted from $1024 \times 1024 \times 3$ to $512 \times 512 \times 3$. Similarly, this method is called Faster R-CNN-512. Undoubtedly, much key information has been lost in the process of downsampling. As a result, the mAP drops from 63.1 to 57.5.

In our proposed method, we feed the selected time-frequency features to Faster R-CNN. When we enable the upsampling in advance, the obtained mAP is 65.1. However, when we carry out wavelet decomposition directly without upsampling, the obtained mAP drops to 60.5. Although this value is lower than the corresponding value for Faster R-CNN-1024, it is still higher than the value for Faster R-CNN-512. More specifically, as shown in the first and the last two rows of Table IV, our method has a 20% performance improvement over DCTNet on all four datasets with comparable algorithm recognition efficiency. In addition, compared to the Faster R-CNN with the same input size, our method also shows approximately 2% performance improvement.

As shown in Fig. 8, we visualize the object detection results in Table IV. These images are selected from the testing dataset of DOTA-v1.0. The detected objects contain various categories, such as airplanes, baseball diamonds, small vehicles, and so on. The objects of each category are represented in a unique color. When we carefully observe the detection results, we can find that though there are many small and dense vehicles, most of them are detected correctly. These vehicles' bounding boxes are tightly wrapped around the objects, and the detected objects are very complete. Therefore, we can qualitatively conclude that the proposed method is very capable according to Fig. 8.

2) Experiments With Our Time-Frequency Analysis Method: Recently, various novel methods have been proposed. Nevertheless, we carefully combed through these articles and handpicked three SOTA methods, including two OBBs-based detectors and one detector for HBBs: DCL, GWD, and SCRDet. What is more, we carry out extensive experiments on various datasets such as DOTA-v1.0, DOTA-v1.5, DOTA-v2.0,

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TABLE IV
COMPARISON WITH OTHER METHODS

method	shape	DOTA-v1.0 HBB mAP	DOTA-v1.5 HBB mAP	DOTA-v2.0 HBB mAP	DIOR	speed (fps)
DCTNet-192	$56 \times 56 \times 192$	43.1	39.3	30.1	43.9	6.2
DCTNet-24	$56 \times 56 \times 24$	44.3	40.1	30.2	45.3	9.4
Faster R-CNN-1024	$1024 \times 1024 \times 3$	63.1	57.4	42.8	64.5	14.3
Faster R-CNN-512	$512 \times 512 \times 3$	57.5	52.4	40.4	58.5	25.0
Our Method-1024	$1024 \times 1024 \times 3$	65.1	59.2	44.3	66.1	7.6
Our Method-512	$512 \times 512 \times 3$	60.5	54.9	44.1	61.7	10.0

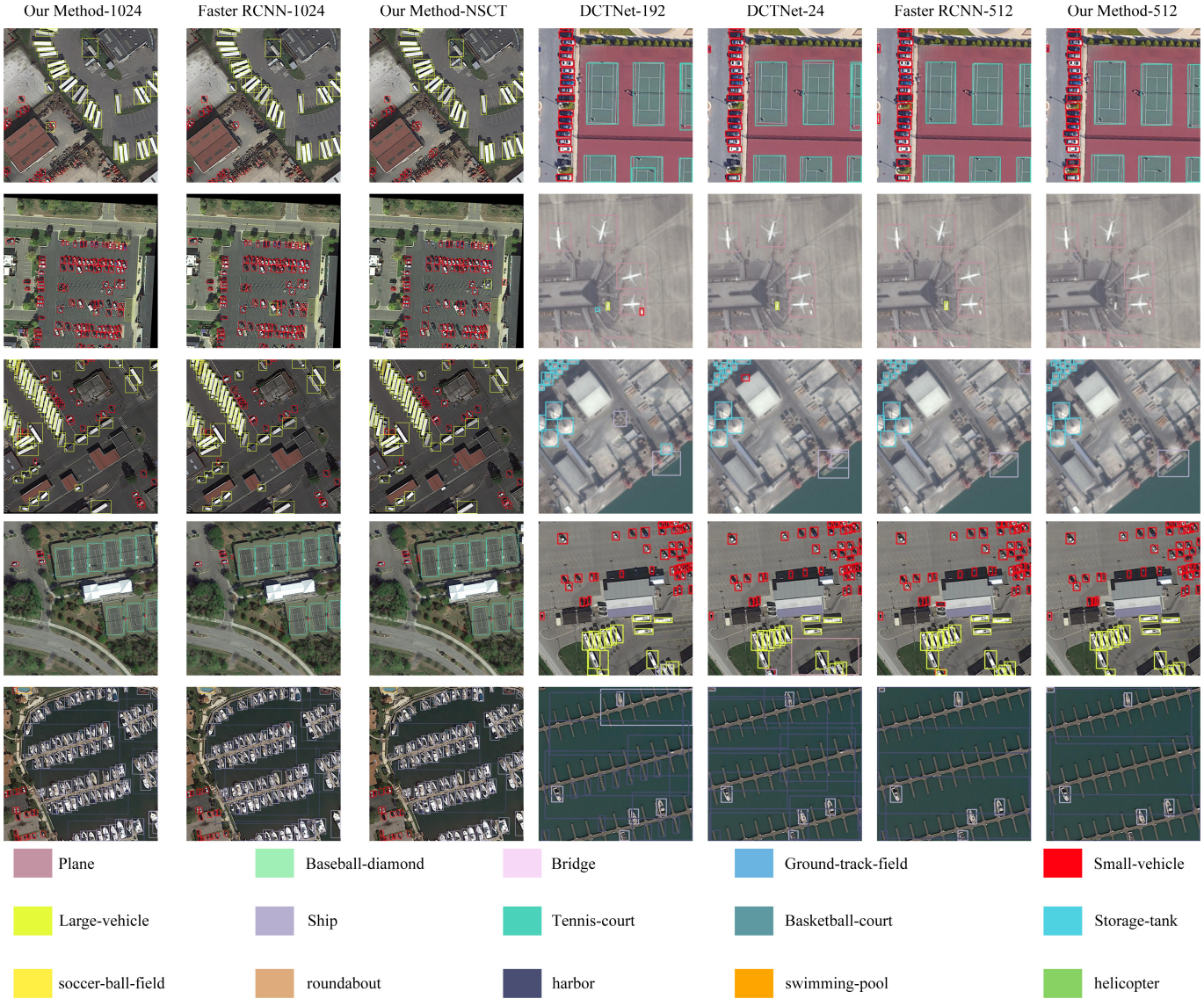


Fig. 8. Visualization of the object detection results on the test dataset of DOTA-v1.0. These images consist of airplanes, baseball diamonds, small vehicles, and so on. The objects of each category are represented in a unique color.

and DIOR. DCL, GWD, and SCRDet are SOTA object detectors. We compare the methods modified by our time-frequency analysis method with the original SOTA methods.

DCL explores a relatively less-studied methodology based on classification. It is designed for angle classification, to replace the sparsely coded label (SCL) in existing classification-based detectors, leading to threefold training speed increase as empirically observed across benchmarks. In addition, it improves the detection accuracy especially for square-like objects, by making DCL-based detectors sensitive to angular distance and object's aspect ratio.

GWD has a novel regression loss based on GWD, which is a fundamental approach to solving boundary discontinuity and inconsistency. GWD can still be informative for learning, even

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TABLE V
COMPARISON WITH SOTA OBB METHODS

method	DOTA-v1.0 OBB mAP	DOTA-v1.5 OBB mAP	DOTA-v2.0 OBB mAP
DCL [57]	67.0	60.2	45.5
Modified DCL (optimal)	68.1	61.1	45.9
GWD [58]	67.9	60.8	46.6
Modified DWG (optimal)	68.8	61.5	47.2

TABLE VI
COMPARISON WITH SOTA HBB METHODS

method	DOTA-v1.0 HBB mAP	DOTA-v1.5 HBB mAP	DOTA-v2.0 HBB mAP	DIOR
SCRDet [59]	74.5	66.1	53.5	71.2
Modified SCRDet (optimal)	76.1	68.0	55.3	73.5

though there is no overlapping between two rotating bounding boxes which is often the case for small object detection. Due to its three unique properties, GWD can also elegantly solve the square-like problem, regardless of how the bounding box is defined.

Since DIOR dataset only has the annotations of HBBs, which does not support the detections of OBBs. As a result, we compare our time-frequency analysis method with DCL and GWD using DOTA-v1.0, DOTA-v1.5, and DOTA-v2.0. Table V shows the comparison results between SOTA OBB methods and methods which are modified by our time-frequency analysis method. All the algorithms in Table V adopt the ResNet-50 with an FPN as backbone. As shown in Table V, after modified by our proposed time-frequency analysis, the mAP of the DCL method improved by 0.9%, 0.9%, and 0.4% for the three DOTA datasets, and the GWD method improved by 0.9%, 0.7%, and 0.6%.

As shown in Fig. 9, we visualize the detection results of the OBB method in Table V. These images are also from the DOTA-v1.0 test dataset. The recognition results for each category are represented by a unique color. In several of the detection results, we can find that the detection bounding box of the modified method has more suitable rotation angle and wraps more completely around the objects. For example, in the last row of Fig. 9, the modified DCL and GWD methods can detect tennis courts more accurately, eliminating the misdetection case before modification.

From Table V and Fig. 9, we can see that our proposed time-frequency analysis is able to further improve the performance of SOTA OBB methods. The above experiments prove that the proposed method is suitable for detecting oriented objects with the outputs of OBBs.

SCRDet is a novel multicategory detector for small, cluttered, and rotated objects, which is made up of a sampling fusion network (SF-Net) and a supervised multidimensional attention network (MDA-Net). Although SCRDet is actually a rotation detector, it can be used for HBB task when we shield the angle parameter in the code. In Table VI, we compare SCRDet with the method modified by our time-frequency analysis method using DOTA-v1.0, DOTA-v1.5, DOTA-v2.0, and DIOR datasets, respectively. As shown in Table VI, the SCRDet modified by our proposed time-frequency analysis achieves approximately 2% performance improvement on all four datasets above. It proves that our proposed time-frequency analysis is also beneficial to the improvement of SOTA HBB method.

TABLE VII
COMPARISON OF COMPUTATIONAL PERFORMANCES

method	speed (fps)
Faster R-CNN	14.3
DCL	9.5
Modified DCL (optimal)	6.0
GWD	10.5
Modified GWD (optimal)	6.5
SCRDet	12.5
Modified SCRDet (optimal)	7.1

D. Computational Complexity Analysis

Deep reinforcement learning usually needs enormous computational resources during training. But this disadvantage was weighed against its merits.

The training time of DDQN is quite time-consuming. When we utilize the whole DOTA training dataset to train the DDQN and the detector together, it costs 200 ms per image with the wavelet subbands shape of $512 \times 512 \times 3$ using one GeForce 1080 GPU. However, in our proposed method, for static feature optimization, the detector carries out forward propagation with $512 \times 512 \times 3$ selected time-frequency channels, which costs 40 ms per image; for dynamic feature optimization with the subbands shape of $512 \times 512 \times 3$, the DDQN and the detector carries out forward propagation together, which costs 100 ms per image. While for the original Faster R-CNN with the input shape of $1024 \times 1024 \times 3$, it costs 70 ms per image during testing. It can be seen that our proposed method is faster than the original Faster R-CNN. The reason for this is that our modified backbone removes the stem convolution layer, which greatly reduces the parameters and computational time.

In summary, the computational time of the proposed method is 40 or 100 ms for static feature optimization and dynamic feature optimization, respectively.

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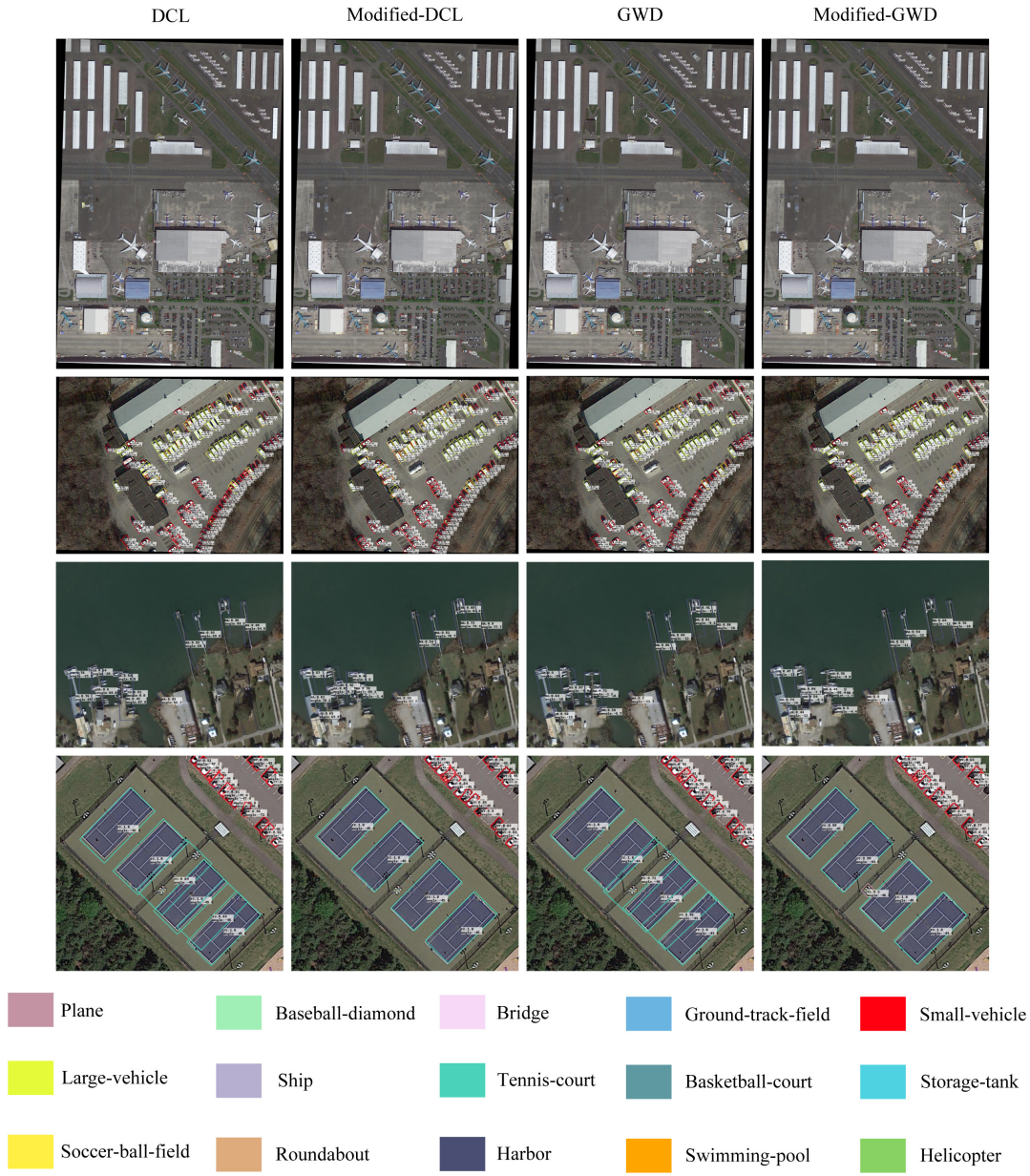


Fig. 9. Visualization of the results between SOTA OBB methods and methods modified by our time-frequency analysis method on the test dataset of DOTA-v1.0. The objects of each category are represented in a unique color. In the last row, the modified DCL and GWD methods can detect tennis courts more accurately, eliminating the misdetection case before modification.

Table VII is the comparison of computational performances. We have compared the computational performances between the proposed method and the other methods used for comparative studies. The input shapes are all $1024 \times 1024 \times 3$ in Table VII.

V. Conclusion

In this article, we develop a robust object detection method for large-scale optical remote-sensing images with complex backgrounds based on time-frequency analysis and feature optimization. With the aim of effectively capturing the time-frequency features, we propose a feature optimization method based on deep reinforcement learning to select the dominant time-frequency channels from the entire time-frequency channels. This procedure reduces computation complexity and ensures the detection performance. In addition, we propose a novel attention mechanism DW-MAM that enables the detector to give more attention to the object area than the background. Inasmuch as DW-MAM is derived from the wavelet function, it can inherently extract the multiscale and multidirection features efficiently, which helps improve the mAP. Many experiments are performed on the DOTA dataset, which shows that the proposed method not only solves the challenges of large scale, but also weakens the interference from complex backgrounds. Our work also proves that the proposed method improves the performance compared

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to the original SOTA object detection methods. In addition, we demonstrate that the proposed method can be used with almost all object detection networks, regardless of whether they are anchor-based or anchor-free detectors, horizontal or rotation detectors.

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